

Akilesh Thuniki

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Professional Summary

DevOps Engineer & Cloud Engineer with over 6+ years of experience designing secure, scalable, and highly available cloud infrastructure on AWS, Azure, and GCP. As an AWS Certified Developer, I lead cloud-native transformations, applying Infrastructure as Code (Terraform) and optimizing CI/CD pipelines for multi-tenant systems. Expert in Kubernetes, ECS, event-driven microservices, and GitOps, with a deep focus on automation, reliability, capacity planning, and cost efficiency. Open source creator of Preapply, a Terraform risk-analysis tool used by engineers in 37 countries and downloaded 2,000+ times alongside SaaSInfraLab and ollama-infra-cli. Reached 99.99% uptime, cut AWS costs by 35%, and accelerated deployments by 70% through resilient design, FinOps, and proactive monitoring (Prometheus, Datadog, Grafana, ELK, Splunk, CloudWatch). Experienced in leading DevSecOps cultural transformation, HIPAA/GDPR/CCPA compliance, complex cloud migrations, incident response, and mentoring high-performing engineering teams, driving cross-functional collaboration, and aligning technology delivery with business strategy and service level agreements.

Technical Skills

- **Cloud Platforms:** AWS (EC2, ECS, EKS, Lambda, S3, SQS, SNS, RDS, VPC, IAM, EBS, ELB, Auto Scaling, Route 53, CloudWatch, CodePipeline), Azure, GCP
- **Infrastructure & Automation:** Terraform (modules, remote state, workspaces), Ansible, Helm, CloudFormation, ArgoCD, Flux, AWS UserData
- **CI/CD & DevOps Tools:** GitHub Actions, GitLab CI, Jenkins, CircleCI, AWS CodePipeline, Chef, Selenium
- **Containerization & Orchestration:** Docker, Docker Compose, Kubernetes, AWS ECS, AWS EKS, OpenShift, KubeFlow, VMware
- **Security & Compliance:** Zero Trust, IAM Policy Management, SonarQube, RBAC, Secrets Management, Shift-Left Security, WAF, Encryption, Audit Logging, HIPAA, GDPR, CCPA
- **Networking:** DNS, TLS/SSL, TCP/IP, CDN, Cloudflare, proxy configuration, network segmentation
- **Monitoring & Observability:** Prometheus, Grafana, ELK Stack, CloudWatch, Splunk, Datadog, AIOps
- **Databases & Data:** DynamoDB, Redis, RDS (PostgreSQL, MySQL), MongoDB, Oracle, BigQuery, Kafka, Kinesis, Spark, EMR, AWS Glue, data pipelines
- **Programming & Scripting:** Python, Java, Bash/Shell, Go, JavaScript, React, YAML, PowerShell
- **Tools & Methodologies:** Git, GitHub, GitLab, Jira, Confluence, ServiceNow, Agile, SaaS, SRE, FinOps, continuous integration, security operations, cross-functional collaboration, Multi-cloud, Hybrid Cloud
- **AI & Developer Tooling:** GitHub Copilot, Cursor, Claude, AI Agents, LLM-assisted workflows, Ollama

Professional Experience

Sr. DevOps Engineer

May 2025 – Present

Client: National Financial Partners, Austin, TX (FinTech / Financial Services)

- Architected a serverless platform on AWS (Lambda, API Gateway, Step Functions) that cut infrastructure overhead by about half and enabled zero-downtime deployments across multi-tenant workloads.
- Established GitOps practices with ArgoCD, GitLab CI/CD, and Terraform, cutting deployment time by 60% and delivering automated integration testing, vulnerability scanning, and security controls across multiple teams in a FinTech environment.
- Implemented DevSecOps automation and security operations across 150+ services, embedding SonarQube, AWS Secrets Manager, and automated 24-hour secret rotation to eliminate hardcoded credentials in alignment with HIPAA and GDPR compliance requirements.
- Engineered a resilient event-driven API ecosystem using SNS, SQS, and Kinesis, improving system reliability by 40% and reducing API latency by 65% through asynchronous streaming architecture.
- Designed a zero-trust security model with TLS encryption, AWS Cognito, WAF, and Shield, achieving 100% compliance with industry security standards and enforcing MFA across all customer-facing applications.
- Optimized AWS spend by right-sizing instances, enabling auto-scaling, and using spot instances, delivering 35% savings while preserving 99.99% uptime, multi-AZ architecture, and SLI/SLO monitoring
- Built serverless ETL/ELT pipelines using AWS Glue and Spark on EMR, with Lambda for orchestration, which increased throughput by 55% and reduced batch-processing latency from 2 hours to under 10 minutes for real-time analytics and ML workloads.
- Established end-to-end observability using CloudWatch, X-Ray, Prometheus, Grafana, ELK, and Splunk, rapid problem solving, reducing incident detection time by 60%, and automating self-healing remediation for 80% of common infrastructure failures via Lambda runbooks.

DevOps Engineer

Feb 2024 – Apr 2025

Client: Gentiva Health, Atlanta, GA (Healthcare / HIPAA-regulated environment)

- Engineered CI/CD pipelines using GitLab CI, GitHub Actions, and AWS ECS/EKS for microservices architecture, improving deployment efficiency by 35% and reducing release failures by 45% through automated testing, rollback mechanisms, and integration test coverage.
- Provisioned cloud infrastructure using Terraform and Ansible with modular templates, reducing environment setup time by 50%, ensuring configuration management consistency, and simplifying onboarding across development teams.
- Engineered scalable event-driven architectures using AWS SQS and Kafka with dead-letter queues, improving system responsiveness by 40% and processing 10M+ streaming events across distributed backend systems.
- Deployed and managed Kubernetes clusters on AWS EKS and OpenShift using Docker Compose and Helm charts, automating container deployments and maintaining fault-tolerant, highly available microservices at scale.
- Partnered with InfoSec to strengthen IAM, RBAC, and audit logging across AWS environments, ensuring full HIPAA, GDPR, and CCPA compliance and reducing unauthorized access risk through performance tuning and SRE practices.
- Migrated legacy monolithic systems to AWS using Terraform and Ansible, refactoring infrastructure as code and improving scalability, reliability, and maintainability of critical healthcare workloads.

DevOps / Build / Release Engineer

Jun 2018 – Jul 2023

Unvired Software India Pvt. Ltd, Hyderabad, India

- Managed end-to-end build and release processes for Java applications across multiple teams, automating CI/CD workflows using Jenkins and Maven, reducing manual release effort by 40% in an Agile/SAFe environment.
- Integrated GitHub with Jenkins using webhooks and build triggers, fully automating application builds, deployments, and integration tests, cutting average build-to-deploy time by 30%.
- Provisioned and managed Linux system administration on AWS EC2 using Puppet and Bash scripts, enabling rapid infrastructure provisioning and reducing environment setup time by 50% through lifecycle automation.
- Acted as Release Manager for two development teams, coordinating cross-team releases, cross-functional stakeholder communication, gathering approvals, and maintaining a zero-missed-deadline record across all production deployments.
- Built and maintained disaster recovery plans for Jenkins and Bamboo, setting up backup servers, reducing CI/CD downtime risk, and ensuring pipeline continuity during failures with defined SLA targets.
- Led incident response and root cause analysis for business-impacting events, identifying recurring failure patterns, implementing preventive fixes, and reducing repeat incidents through continuous improvement.

Certifications

- AWS Certified Developer – Associate
- AWS Cloud Practitioner Essentials — Amazon Web Services
- Agile Software Development — University of Minnesota
- QA Automation Engineer

Open-Source Projects

Preapply — Terraform Risk Analysis Tool

<https://pypi.org/project/preapply>

Python, Terraform, Ollama, PyPI, GitHub Actions

github.com/akileshthuniki

- Designed and published a deterministic Terraform plan risk analyzer that computes blast radius, scores deployment risk, and explains infrastructure impact, enabling DevOps teams to catch failures before production through shift-left validation and CI best practices.
- Adopted by engineers in 37 countries with 2,074+ verified downloads (Google BigQuery / PyPI dataset), including enterprise adoption via organizational package mirroring (bandersnatch) and Nexus repositories. Peak adoption: 1,943 downloads in February 2026 across macOS, Linux, and Windows on Python 3.8–3.13.
- Integrated an optional local AI advisory layer (Ollama) providing human-readable explanations without compromising deterministic analysis, demonstrating responsible AI usage in safety-critical infrastructure systems.
- Published under Apache-2.0 license; version 1.0.3 on PyPI with full CI/CD via GitHub Actions, test coverage, and contributor documentation.

AWS EKS, Terraform, Kubernetes, GitOps, GitHub Actions

- Built a fully automated multi-tenant SaaS infrastructure on AWS using EKS, Terraform, and GitOps, solving tenant isolation, automated onboarding/offboarding, and cost governance at scale; applicable to fintech, healthcare, and enterprise SaaS platforms.
- Implemented strict per-tenant isolation using IAM, IRSA, Kubernetes RBAC, namespace quotas, and NetworkPolicies, demonstrating production-grade security architecture for regulated industries.
- Published architecture, code, and documentation publicly for community reuse; featured in 2 technical blog posts on Medium, Dev. to, and Hashnode covering multi-tenant infrastructure patterns.

ollama-infra-cli

github.com/akileshthuniki/ollama-infra-cli

Python, Ollama, CLI, DevOps Automation

- Developed a lightweight CLI for interacting with local Ollama AI models in DevOps workflows — supporting model listing, stdin-based prompt execution, and verbose debugging mode for infrastructure automation and LLM-assisted scripting without cloud API dependencies.

Publications & Thought Leadership

- Published 3 technical articles on Medium, Dev.to, and Hashnode covering multi-tenant SaaS infrastructure on AWS, GitOps automation, and responsible AI in DevOps workflows (Oct–Feb 2026).

Education

Master of Science in Information Systems

Aug 2024

University of Utah, Salt Lake City, USA

Bachelor of Technology in Computer Science & Engineering

Jun 2019

Jawaharlal Nehru Technological University, Hyderabad, India